

**VALIDATION REPORT:
PHOTON ATTENUATION SIMULATOR
(V1.07)**

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1. EXECUTIVE SUMMARY

This report documents the validation of the Shielding Attenuation Simulator V1.07. Version 1.07 adds target classes, minimum shielding thickness calculations, and buildup-aware design behavior for single-material shielding cases.

The baseline attenuation model, NIST XCOM interpolation, point source inverse-square model, isotope source model, and Geometric Progression (G-P) buildup equation are unchanged from the previously validated V1.06 implementation. Therefore, this report focuses only on the new target-driven minimum thickness calculator pathway.

V1.07 supports three target types:

- Transmission target
- Reduction factor target
- Flux target

For manual monoenergetic photon sources, V1.07 supports an analytical narrow-beam thickness solution. For isotope sources, V1.07 uses bisection because each photon line attenuates differently and the total source flux is calculated by summing line-by-line results.

V1.07 also adds optional buildup aware minimum thickness design. When buildup is requested, the simulator attempts to solve the design problem using buildup-corrected flux. If the buildup model is unsupported or cannot reach the target within the valid G-P buildup range, the simulator falls back to narrow-beam design and reports a warning instead of extrapolating buildup factors.

2. METHODOLOGY AND PHYSICS MODELS

2.1 Previously Validated Attenuation, Source, and Buildup Models

The narrow-beam attenuation model remains unchanged. Shield transmission is still calculated using the Beer-Lambert law, with NIST XCOM mass attenuation coefficients converted to linear attenuation coefficients using material density. Detector flux is still calculated using isotropic point-source geometric spreading.

The isotope source model introduced in V1.06 also remains unchanged. Isotope sources are still processed line-by-line, with each photon line receiving its own attenuation calculation and source-level flux contribution.

The G-P buildup calculation also remains based on the V1.05 and V1.06 implementation. V1.07 does not add new G-P coefficient tables, a new buildup factor equation, or a multilayer buildup treatment. The new behavior is the use of existing buildup calculations in a target-driven minimum-thickness design mode.

2.2 Target Model

The target model converts each supported target into an equivalent detector flux requirement. This allows the minimum thickness calculator to use a common comparison condition:

$$final\ flux \leq target\ flux$$

The three target conversions are:

- TransmissionTarget: $\phi_{target} = \phi_{unshielded} \cdot T_{target}$
- ReductionFactorTarget: $\phi_{target} = \frac{\phi_{unshielded}}{Reduction\ Factor}$
- FluxTarget: ϕ_{target} is supplied directly by the user

where:

- ϕ is flux in photons/cm²/s.
- T is the transmission fraction.

2.3 Manual Source Minimum Thickness

For manual monoenergetic photon sources in narrow-beam mode, the simulator uses the analytical attenuation solution.

$$T = \frac{\phi_{target}}{\phi_{unshielded}}$$

$$x = - \frac{\ln(T)}{\mu}$$

where:

- x is the required shielding thickness in cm.
- μ is the linear attenuation coefficient in cm⁻¹.

This test pathway validates that the simulator can directly solve for required thickness without changing the underlying attenuation model.

2.4 Isotope Source Minimum Thickness

For isotope sources, the simulator uses bisection over shield thickness. This is necessary because isotope sources may contain multiple photon lines, and each line has a different attenuation coefficient. At each trial thickness, the simulator calculates all line-by-line flux values and sums the total source flux.

The bisection search continues until the smallest thickness within tolerance is found such that the total detector flux satisfies the requested target.

2.5 Buildup-Aware Design and Fallback Behavior

When buildup is requested in minimum-thickness mode, V1.07 evaluates the target using buildup-corrected flux. Since the buildup factor depends on shield thickness in mean free paths, buildup-aware design uses bisection instead of the analytical narrow-beam formula.

For isotope sources, every photon line must receive a valid buildup correction for a buildup-aware total to be reported. If any line cannot use G-P buildup, the buildup-aware solution is considered unavailable.

The simulator does not extrapolate G-P buildup beyond the supported 40 MFP range. If the buildup-aware solution cannot meet the target within the valid G-P buildup thickness range, the simulator reports a warning and falls back to narrow-beam minimum-thickness design.

3. TEST CASE RESULTS

3.1 Narrow-Beam Minimum Thickness Tests

The validation runner checks the new target driven minimum thickness calculation pathway using manual and isotope sources. These tests validate target conversion, analytical manual source thickness calculation, isotope source bisection, no-shielding behavior, and impossible target rejection.

Test Case	Input Parameters	Expected Result	Actual Result	Status
1	Manual source, lead, TransmissionTarget(0.1)	Final transmission = 0.1	Final transmission = 0.1	PASS
2	Manual source, lead, ReductionFactorTarget(1000)	Final transmission = 0.001	Final transmission = 0.001	PASS
3	Manual source, lead, FluxTarget(100 photons/cm ² /s)	Final flux <= 100 photons/cm ² /s	Final flux <= 100 photons/cm ² /s	PASS
4	Cs-137 isotope source, lead, FluxTarget(100 photons/cm ² /s)	Final flux <= 100 photons/cm ² /s	Final flux <= 100 photons/cm ² /s	PASS
5	Cs-137 isotope source, lead	Required thickness > 0	Required thickness > 0	PASS
6	Manual source, very high flux target	Required thickness = 0 warning generated	Required thickness = 0 warning generated	PASS

7	Manual source, impossible target with max thickness = 0.1 cm	ValueError	ValueError	PASS
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3.2 Buildup-Aware Minimum Thickness Tests

The validation runner verifies that buildup-aware design can use G-P buildup when valid and can fall back to narrow-beam design when buildup is unavailable. The manual buildup-aware thickness is also checked against the narrow-beam thickness to confirm that buildup produces a more conservative design thickness.

Test Case	Input Parameters	Expected Result	Actual Result	Status
8	Manual source, lead, FluxTarget(100), buildup enabled	Final buildup-corrected flux ≤ 100	Final buildup-corrected flux ≤ 100	PASS
9	Manual source, buildup enabled	buildup_used = True	buildup_used = True	PASS
10	Manual buildup-aware result vs narrow-beam result	Buildup-aware thickness > narrow-beam thickness	Buildup-aware thickness > narrow-beam thickness	PASS
11	Cs-137 isotope source, lead, FluxTarget(100), buildup enabled	Final buildup-corrected flux ≤ 100	Final buildup-corrected flux ≤ 100	PASS
12	Cs-137 isotope source, buildup enabled	buildup_used = True	buildup_used = True	PASS
13	Manual 0.059541 MeV source, FluxTarget($1 \cdot 10^{-121}$), buildup enabled	Fallback result reaches flux target	Fallback result reaches flux target	PASS
14	Manual 0.059541 MeV source, buildup enabled	buildup_used = False	buildup_used = False	PASS
15	Manual 0.059541 MeV source, buildup enabled	Warning generated	Warning generated	PASS

3.3 Manual Command-Line Example Tests

Manual command-line tests were performed to confirm that the new minimum-thickness design mode produces readable user-facing output and clearly distinguishes buildup-aware results from fallback narrow-beam results.

Test Case	Input Parameters	Expected Result	Actual Result	Status
16	Cs-137, 1 Ci, lead, $r = 100$ cm, FluxTarget(100), buildup enabled	Buildup-aware minimum thickness design	Buildup-aware minimum thickness design	PASS
17	Manual source 0.059541 MeV, $S = 3.7 \cdot 10^{10}$ photons/s, lead, $r = 100$ cm, FluxTarget($1 \cdot 10^{-121}$), buildup enabled	Fallback to narrow-beam with warning	Fallback to narrow-beam with warning	PASS

3.4 Manual Source Regression Test

This confirms that the V1.07 target and minimum-thickness additions did not alter the validated single-energy calculation pathway.

Test Case	Input Parameters	Expected Result	Actual Result	Status
18	0.6617 MeV, $S = 3.7 \cdot 10^{10}$ photons/s, 5 cm lead, $r = 100$ cm	$\phi_{\text{uncollided}} \approx 5.40662 \cdot 10^2$ photons/cm ² /s	$\phi_{\text{uncollided}} \approx 5.40662 \cdot 10^2$ photons/cm ² /s	PASS
19	0.6617 MeV, $S = 3.7 \cdot 10^{10}$ photons/s, 5 cm lead, $r = 100$ cm, buildup enabled	$\phi_{\text{buildup}} \approx 1.42612 \cdot 10^3$ photons/cm ² /s	$\phi_{\text{buildup}} \approx 1.42612 \cdot 10^3$ photons/cm ² /s	PASS

The Cs-137 example demonstrates that buildup-aware design requires more shielding than a narrow-beam-only comparison because buildup-corrected flux includes scattered photon contribution. The low-energy manual source example demonstrates that V1.07 does not extrapolate G-P buildup outside the valid range and instead reports a narrow-beam fallback result with a warning.

4. ASSUMPTIONS AND LIMITATIONS

1. Minimum-Thickness Mode Uses One Material: V1.07 minimum thickness design selects the required thickness for one shielding material at a time.

2. Single Layer Homogeneous Buildup Only: G-P buildup correction remains supported only for single-layer homogeneous shielding cases.
3. Buildup-Aware Design is Single-Layer Only: Buildup-aware minimum-thickness design is limited to single-layer homogeneous shielding because the implemented G-P buildup model is single-layer only.
4. No Multilayer Optimization: V1.07 does not optimize multilayer shielding stacks. Multilayer optimization is reserved for a later version.
5. No Multilayer Buildup Treatment: Multilayer buildup is not implemented. The simulator rejects buildup mode for multiple shielding layers because applying a single material buildup factor to mixed materials may produce misleading results.
6. Supported Buildup Materials: V1.07 supports G-P exposure buildup correction for lead, aluminum, ordinary concrete, water, tungsten, copper, and tin.
7. Unsupported Buildup Materials: Barite concrete and polyethylene remain unsupported for buildup correction in V1.07, although they remain available for narrow-beam attenuation calculations.
8. Point-Source Geometry: The source is still modeled as an isotropic point source.
9. Narrow-Beam Transmission Still Used as Baseline: The buildup-corrected flux is calculated by multiplying the narrow-beam uncollided flux by the G-P buildup factor.
10. Coefficient Interpolation: G-P coefficients are linearly interpolated between tabulated energies. This is separate from the NIST XCOM attenuation interpolation, which uses log-log interpolation.
11. Exposure Buildup Coefficients: The implemented coefficients are exposure buildup coefficients. Future versions may distinguish exposure, dose, and energy buildup factors if additional coefficient tables are added.
12. Simple 1D Geometry: The simulator assumes shielding thickness lies along the source to detector path. Therefore, total shielding thickness cannot exceed detector distance.
13. Selected Photon Lines: The isotope library introduced in V1.06 uses selected major photon lines, not complete emitted decay spectra.
14. Emitted Photons Before Source Self-Attenuation: Photon intensities are treated as emitted photons per decay before source encapsulation, source self-attenuation, air attenuation, or detector-window effects.
15. No Dose Conversion: V1.07 reports photon flux, not dose rate, exposure rate, air kerma, or effective dose.
16. G-P Range Limit: Buildup correction is only applied through 40 mean free paths. If a photon line exceeds this range, narrow-beam flux is reported but buildup is skipped with a warning.

5. CONCLUSION

The Shielding Attenuation Simulator V1.07 successfully adds target-driven minimum shielding thickness calculations while preserving the previously validated attenuation, source, and buildup physics models.

Validation confirmed that transmission targets, reduction-factor targets, and flux targets are converted correctly into detector flux requirements. Manual monoenergetic sources use the analytical narrow-beam attenuation solution, while isotope sources use bisection to handle line-by-line attenuation and source-level flux summation.

The V1.07 buildup-aware design pathway was also validated. When G-P buildup is valid, the simulator can calculate a buildup-aware required thickness and correctly reports buildup-corrected flux as the target-comparison quantity. When buildup is unavailable or cannot meet the target within the valid G-P mean-free-path range, the simulator falls back to narrow-beam design and reports a warning.

V1.07 represents a practical improvement over V1.06 by allowing the simulator to answer design questions such as how much shielding is required to meet a specified transmission, reduction-factor, or flux target. Future work should focus on material comparison, constrained optimization using cost and mass, plotting, scenario studies, GUI development, and eventual comparison against Monte Carlo transport tools.

References

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